

NORWEGIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

IT3010 - 1A

RESEARCH-BASED INNOVATION METHODOLOGIES IN COMPUTER AND INFORMATION
SCIENCE

Analyzing the impact of COVID-19 on e-learning technologies

Group members

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Group number: 13

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<https://www.ntnu.edu/studies/courses/IT3010>



Norwegian University of
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Motivation

In 2020 the whole world underwent an extremely quick change in how they led their everyday lives. Education went from being mostly in-person to then primarily online. Even though the World Health Organization still classifies covid as a public health emergency, most countries have removed most if not all restrictions that were set [3, 1]. The target of our research, NTNU, lifted most of its restrictions around spring of 2022. With the world opening again, there is uncertainty regarding whether or not education can return to its "normal" ways. Before the pandemic learning was traditionally done in a classroom with a lecturer presenting information and students listening. There had been some development on e-learning, but this was not used or tested on a broader scale [5]. During covid, most educational institutions went completely to using e-learning in form of Microsoft teams or ZOOM. The structural method of set lecture times held by professional teachers remained through a digital medium [2, 5]. Now that most things have gone back to normal, it begs the question of whether or not the recent events have made any permanent impact on how we see learning. There might be valuable takeaways from how things were done during the Covid outbreaks.

E-learning has proven beneficial in times when meeting in person has been difficult. This has given everyone that has the technological ability and access to learn regardless of their background, disability, or quality of life [4]. When looking for how educational institutions continued to use these learning technologies in the forms of online videos, lectures and communicative platforms, there was a lack of research on how these e-tools were to be used going forward. Where it is possible, there should be a focus on having a hybrid school system, where resources should be distributed to give everyone equal opportunity [5].

Research Objective

The research objective is to get an understanding of how students and professors at NTNU want the future of education to look like and what technologies are a part of that future. This is in hopes of finding a compromise that most people can agree with.

- The current research is location based around the world, how did it effect NTNU.
- research area cooperation and learning technologies
- Why the changes made during covid didn't stay, even if some of them were beneficiary?
- Focus the scope on Norway (Trondheim)

Research Questions

- How has covid impacted the usage of e-learning technologies at NTNU post-covid?
- How was e-learning technologies under covid?
- Did everything go back to how it was pre-covid?
- What do the students feel about it?
- What's the view from the professors regarding adapting changes?

References

- [1] The World Tourism Organization (UNWTO). *COVID-19 Related Travel Restrictions*. 2023-. URL: <https://www.unwto.org/covid-19-travel-restrictions>. (accessed: 09.02.2023).
- [2] Sarah I. Hofer, Nicolae Nistor, and Christian Scheibenzuber. “Online teaching and learning in higher education: Lessons learned in crisis situations”. In: *Computers in Human Behavior* 121 (2021), p. 106789. ISSN: 0747-5632. DOI: <https://doi.org/10.1016/j.chb.2021.106789>. URL: <https://www.sciencedirect.com/science/article/pii/S0747563221001126>.
- [3] World Health Organization. *Coronavirus disease (COVID-19) Weekly Epidemiological Updates and Monthly Operational Updates*. URL: <https://www.who.int/emergencies/diseases/novel-coronavirus-2019/situation-reports>. (accessed: 09.02.2023).
- [4] Ping Qiao et al. “The Development and Adoption of Online Learning in Pre- and Post-COVID-19: Combination of Technological System Evolution Theory and Unified Theory of Acceptance and Use of Technology”. In: *Journal of Risk and Financial Management* 14.4 (2021). ISSN: 1911-8074. DOI: [10.3390/jrfm14040162](https://doi.org/10.3390/jrfm14040162). URL: <https://www.mdpi.com/1911-8074/14/4/162>.
- [5] Yong Zhao and Jim Watterston. “The changes we need: Education post COVID-19”. In: *Journal of Educational Change* 22.1 (2021), pp. 3–12. DOI: <https://doi.org/10.1007/s10833-021-09417-3>.

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Meldeskjema

Referansenummer

284624

Hvilke personopplysninger skal du behandle?

- Lydopptak av personer
- Bakgrunnsopplysninger som vil kunne identifisere en person
- Helseopplysninger

Beskriv hvilke bakgrunnsopplysninger du skal behandle

Bostedskommune, arbeidssted, stilling

Prosjektinformasjon

Prosjekttittel

IT3010 - group 13

Prosjektbeskrivelse

Analyzing the impact of COVID-19 on e-learning technologies

Begrunn hvorfor det er nødvendig å behandle personopplysningene

Det som nødvendig for å realisere formålet med analysen.

Ekstern finansiering

Ikke utfyllt

Type prosjekt

Studentprosjekt, masterstudium

Kontaktinformasjon, student

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Behandlingsansvar

Behandlingsansvarlig institusjon

Norges teknisk-naturvitenskapelige universitet / Fakultet for informasjonsteknologi og elektroteknikk (IE) / Institutt for datateknologi og informatikk

Prosjektansvarlig (vitenskapelig ansatt/veileder eller stipendiat)

-

Skal behandlingsansvaret deles med andre institusjoner (felles behandlingsansvarlige)?

Nei

Utvalg 1

Beskriv utvalget

Studenter, ansatte NTNU

Beskriv hvordan rekruttering eller trekking av utvalget skjer

Tilfeldig i grupper som har opplevd effekten av COVID-19 på universitetsnivå.

Alder

18 - 70

Personopplysninger for utvalg 1

- Lydopptak av personer
- Bakgrunnsopplysninger som vil kunne identifisere en person
- Helseopplysninger

Hvordan samler du inn data fra utvalg 1?

Personlig intervju

Grunnlag for å behandle alminnelige kategorier av personopplysninger

-

Grunnlag for å behandle særlige kategorier av personopplysninger

Elektronisk spørreskjema

Grunnlag for å behandle alminnelige kategorier av personopplysninger

-

Grunnlag for å behandle særlige kategorier av personopplysninger

Papirbasert spørreskjema

Grunnlag for å behandle alminnelige kategorier av personopplysninger

-

Grunnlag for å behandle særlige kategorier av personopplysninger

Informasjon for utvalg 1

Informerer du utvalget om behandlingen av personopplysningene?

Ja

Hvordan?

Skriftlig informasjon (papir eller elektronisk)

Tredjepersoner

Skal du behandle personopplysninger om tredjepersoner?

Nei

Dokumentasjon

Hvordan kan de registrerte få innsyn, rettet eller slettet personopplysninger om seg selv?

De kan spørre om det.

Totalt antall registrerte i prosjektet

1-99

Tillatelser

Skal du innhente følgende godkjenninger eller tillatelser for prosjektet?

Ikke utfyllt

Behandling

Hvor behandles personopplysningene?

- Private enheter

Hvem behandler/har tilgang til personopplysningene?

- Prosjektansvarlig
- Student (studentprosjekt)

Tilgjengeliggjøres personopplysningene utenfor EU/EØS til en tredjestat eller internasjonal organisasjon?

Nei

Sikkerhet

Oppbevares personopplysningene atskilt fra øvrige data (koblingsnøkkel)?

Nei

Begrunn hvorfor personopplysningene oppbevares sammen med de øvrige opplysningene**Hvilke tekniske og fysiske tiltak sikrer personopplysningene?**

- Personopplysningene anonymiseres fortløpende
- Adgangsbegrensning

Varighet

Prosjektperiode

09.02.2023 - 05.05.2023

Hva skjer med dataene ved prosjektslutt?

Data slettes (sletter rådataene)

Vil de registrerte kunne identifiseres (direkte eller indirekte) i oppgave/avhandling/øvrige publikasjoner fra prosjektet?

Nei

Tilleggsopplysninger
